

GRANTLEY ADAMS INTL, Barbados

WMO: 789540

Lat: 13.0853N

Lon: 59.4873W

Elev: 217

StdP: 14.58

Time Zone: -4.00 (W04)

Period: 94-19

WBAN: 00408

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	72.8	73.5	64.2	90.6	77.8	65.8	96.0	78.1	28.5	79.4	27.3	79.8	9.0	50	0.603

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9	9.0	88.2	79.4	88.1	79.4	87.8	79.3	81.2	85.7	80.8	85.6	80.3	85.3	15.5	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
80.3	158.3	84.2	79.2	152.7	83.6	79.0	151.7	83.6	45.1	86.2	44.5	86.4	44.1	85.6	85.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 25.9	24.3	22.8	69.4	89.6	1.5	0.8	68.4	90.2	67.5	90.6	66.7	91.1	65.6	91.6		
(5)			65.8	83.1	1.5	1.2	64.7	83.9	63.9	84.6	63.0	85.2	61.9	86.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	81.5	79.5	79.4	80.1	81.2	82.5	82.7	82.5	82.7	82.7	82.4	81.7	80.6
	DBStd	1.95	1.40	1.50	1.61	1.59	1.37	1.32	1.28	1.52	1.75	1.62	1.55	1.44
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	11497	914	822	933	937	1008	980	1007	1014	981	1003	950	948
	CDD65	6022	449	402	468	487	543	530	542	549	531	538	500	483
	CDH74	62096	3748	3261	4093	4795	5938	5882	6046	6288	6146	6035	5275	4589
	CDH80	16875	593	555	854	1183	1749	1734	1777	2061	2120	1914	1413	922
(14) Wind	WSAvg	14.2	15.0	15.6	14.7	14.6	15.3	16.8	15.2	12.7	10.9	11.8	12.5	14.8
(15) Precipitation	PrecAvg	46.0	2.6	1.7	1.5	2.3	2.1	4.1	4.9	5.9	5.9	6.7	6.1	4.3
	PrecMax	73.0	7.9	4.1	3.7	12.0	8.1	9.4	13.0	13.2	14.7	18.0	18.0	11.0
	PrecMin	33.2	0.4	0.1	0.3	0.2	0.4	0.3	1.1	1.5	2.0	1.4	1.4	0.5
	PrecStd	8.2	1.5	0.9	0.8	2.1	1.9	2.2	2.6	2.4	3.0	3.3	3.7	2.6
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	86.1	86.2	87.6	88.0	88.2	88.2	88.2	89.1	89.7	89.5	88.1	86.4
		MCWB	76.3	76.5	77.2	78.5	78.9	78.6	79.0	79.7	80.2	80.1	80.0	77.6
	2%	DB	84.6	84.6	86.1	86.4	87.9	87.8	87.7	88.1	88.2	88.1	86.8	85.9
		MCWB	75.6	75.1	76.2	77.0	78.5	78.6	78.9	79.6	79.8	79.9	79.2	77.5
	5%	DB	84.1	84.2	84.7	86.1	86.8	86.4	86.4	87.7	87.9	87.6	86.2	84.6
		MCWB	75.4	74.7	75.3	76.8	77.9	78.2	78.6	79.6	79.7	79.7	78.8	76.6
	10%	DB	82.8	83.2	84.2	85.4	86.1	86.1	86.4	87.4	87.4	86.3	85.8	84.3
		MCWB	74.7	74.2	75.1	76.4	77.5	78.1	78.5	79.0	79.5	79.1	78.6	76.3
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	78.6	78.4	78.7	79.8	81.3	81.0	81.3	81.8	81.8	81.7	81.2	79.8
		MCDB	82.9	84.0	84.6	85.3	85.7	85.0	85.2	86.0	86.6	86.1	85.4	83.8
	2%	WB	77.4	77.0	77.6	78.8	80.2	80.3	80.5	80.9	80.9	80.8	80.5	78.9
		MCDB	82.3	82.4	83.6	84.1	85.0	84.7	85.0	85.8	86.2	85.9	85.0	83.2
	5%	WB	76.6	76.2	76.8	78.1	79.3	79.5	79.8	80.3	80.4	80.2	79.7	78.2
		MCDB	81.6	81.6	82.8	83.5	84.2	84.0	84.3	85.4	85.8	85.3	84.2	82.7
	10%	WB	76.0	75.4	76.1	77.5	78.7	79.0	79.1	79.7	79.8	79.6	79.1	77.6
		MCDB	81.0	80.9	82.0	82.9	83.7	83.6	84.6	85.1	85.4	84.5	83.5	82.2
(35) Mean Daily Temperature Range	5% DB	MDBR	8.4	8.8	8.9	8.7	8.0	7.7	8.0	8.6	9.0	9.0	8.5	8.2
		MCDBR	8.8	9.0	9.1	8.9	8.3	8.0	8.4	9.0	9.5	9.5	8.8	8.4
		MCWBR	4.8	4.8	4.7	4.4	4.3	4.1	4.3	4.6	4.9	5.0	4.8	4.7
	5% WB	MCDBR	8.2	8.5	8.6	8.2	7.8	7.6	8.0	8.6	9.0	9.1	8.5	8.0
		MCWBR	4.9	5.1	4.8	4.4	4.5	4.4	4.6	4.9	5.0	5.1	4.9	4.8
(40) Clear-Sky Solar Irradiance	taub		0.376	0.393	0.390	0.434	0.477	0.561	0.519	0.459	0.422	0.385	0.379	0.385
	taud		2.575	2.497	2.524	2.406	2.263	2.002	2.135	2.367	2.488	2.625	2.625	2.572
	Ebn at Noon		294	292	294	278	262	238	249	268	280	291	291	288
	Edh at Noon		31	35	35	39	44	56	49	40	35	30	30	31
(44) All-Sky Solar Radiation	RadAvg		1696	1916	2044	2087	2036	1941	1992	1995	1935	1754	1612	1615
	RadStd		76	48	89	82	116	81	98	101	106	74	117	79

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page